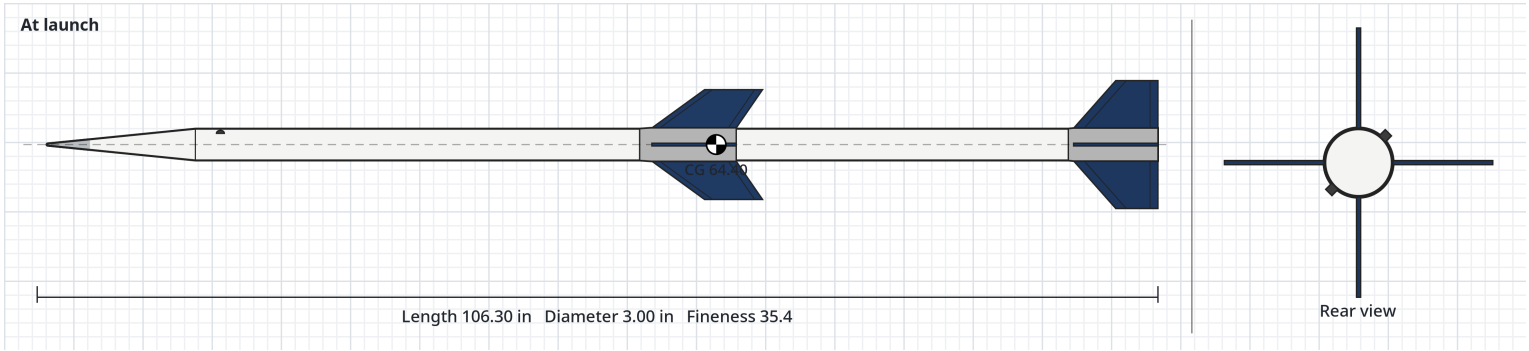
PANDORA - EAYF_CONFIG_1 - Results



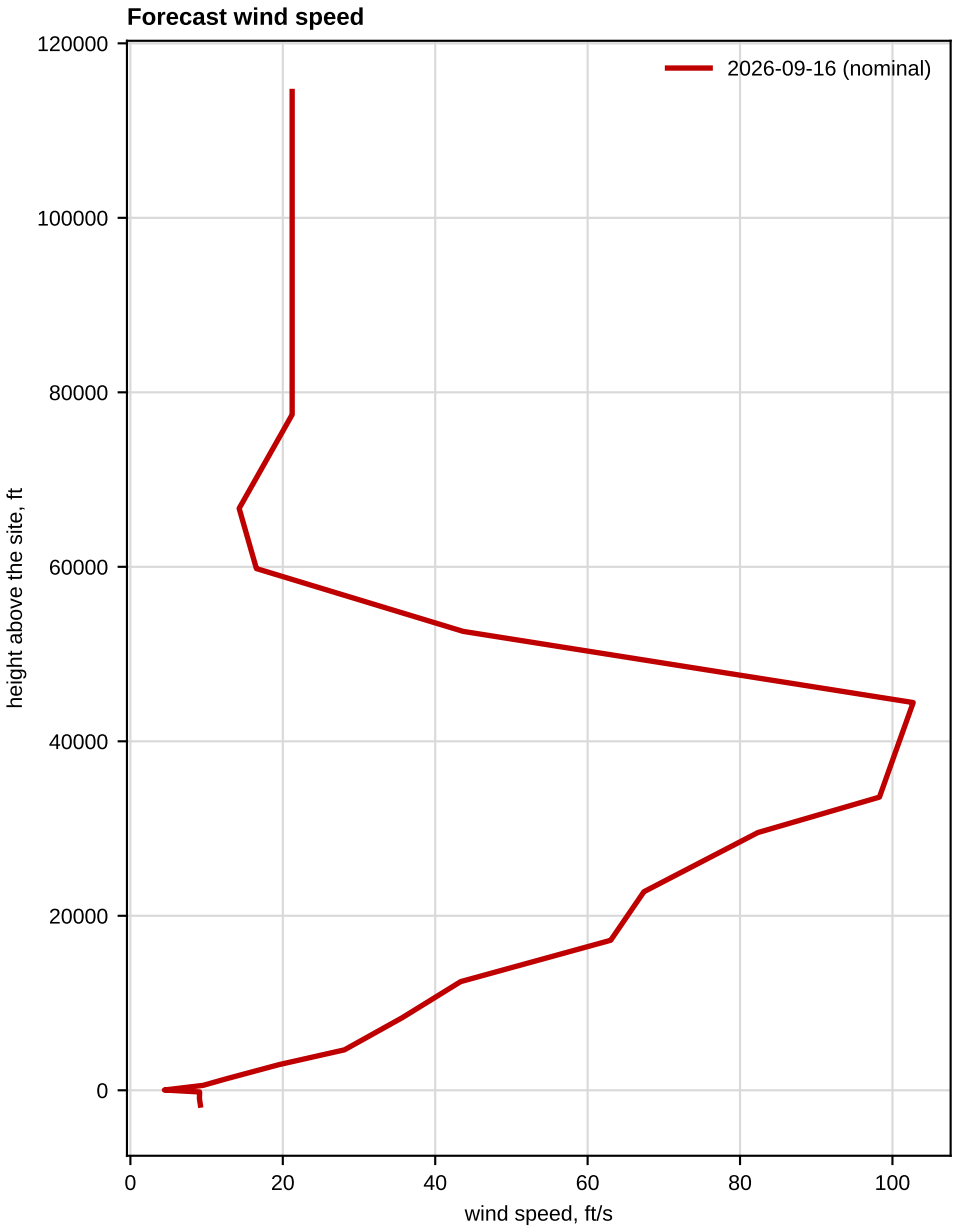
PANDORA - EAYF_CONFIG_1
Nose radius for heating: 0.1 in

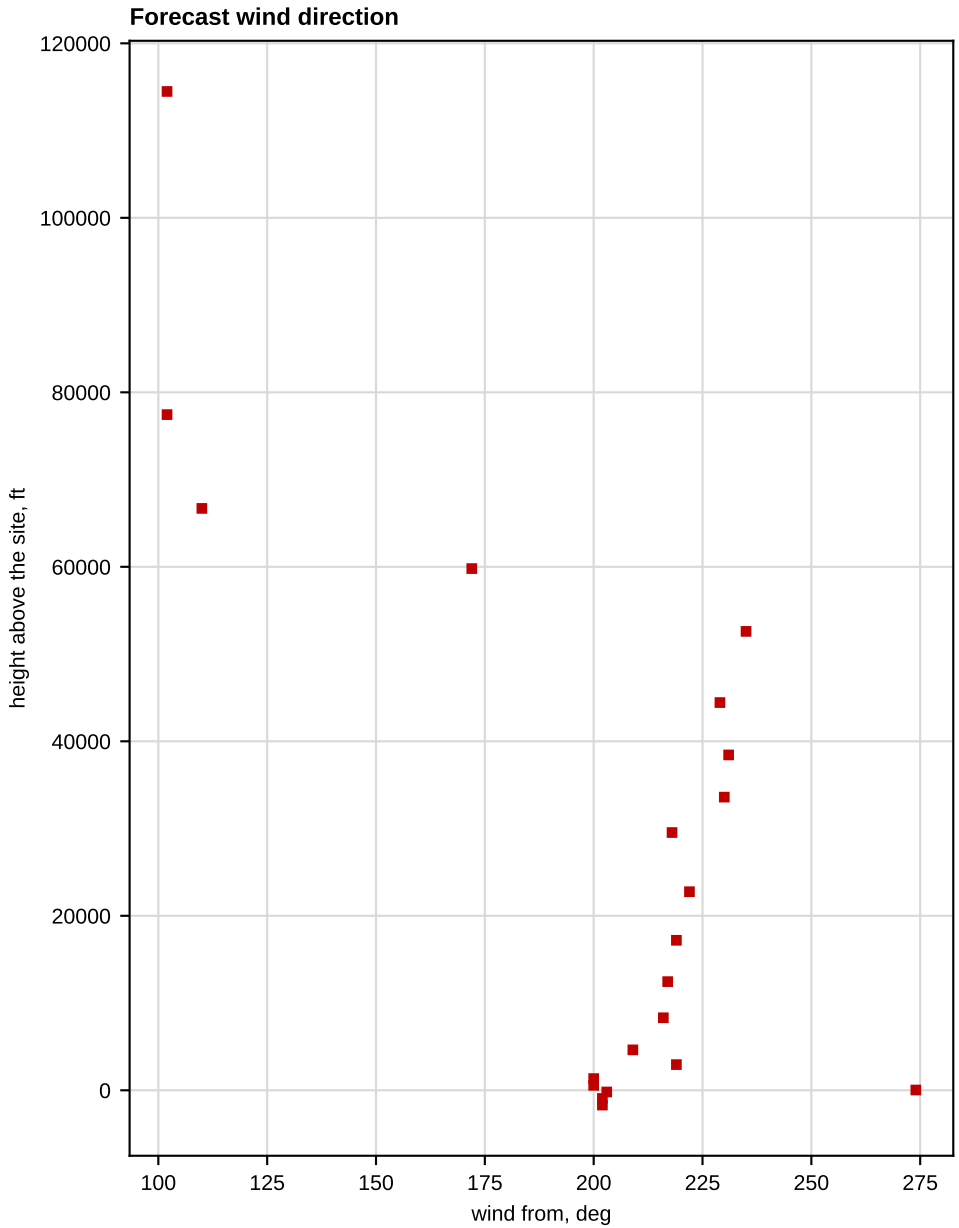
	nominal	
Rail exit speed	114.08	ft/s
Rail exit AoA	2.2067	deg
Forecast gust	13.123	ft/s
Rail exit AoA in gust	6.5623	deg
Apogee	1.1778e+05	ft
Time to apogee	88.739	s
Max speed	4354.1	ft/s
Max acceleration	73.846	g
Max roll rate	7.3392	rev/s
Max angle of attack	1.2518	deg
Max AoA after burnout	0.39259	deg
Roll-pitch crossings	0	
Resonance time	nan	s
Resonance altitude	nan	ft
AoA rise at resonance	nan	deg
Roll lock-in held	0	s
Lock-in starts	nan	s
Max Mach	4.1188	
Max dynamic pressure	11225	psf
Peak vibration, thrust	25.234	Grms
Peak thrust vibration at	6.3	s
Peak vibration, lateral	25.234	Grms
Peak lateral vibration at	6.3	s
Peak heat flux	554.05	BTU/ft²/s
Heat load	2971.1	BTU/ft²
Tilt at ignition	3.9992	deg
Altitude at ignition	12050	ft
Flight time	369.34	s
Landing east	-1.6372	mi
Landing north	-1.3332	mi
Landing distance	2.1114	mi
Impact speed	40.683	ft/s
Launch mass	38.822	lb, CG 64.40 in from the nose tip
Min static margin	3.1301	cal (while loaded)

Angles of attack are while loaded, launch to apogee: dynamic pressure above 5% of its peak.
Failure conditions: lock-in = true; resonance = true; lockout = true. The nominal flight passes
Sustainer ignited: yes
Launch cylinder, 7 nmi: the nominal flight lands inside

PANDORA - EAYF_CONFIG_1 - Weather

```
Weather
  Open-Meteo forecast at 35.3471, -117.8082, 17:00 UTC
  day                   surface wind         gust at the launch hour
  2026-09-16      9.2 ft/s from 202 deg   13.1 ft/s from 274 deg at 17:00   nominal flight
  Each day is a whole forecast wind profile; the gust is the 10 m forecast for the launch hour, used only for the rail exit AoA in gust.
```





PANDORA - EAYF_CONFIG_1 - Inputs

```
[Case]
  Units               IN
  Analyze             STACK

[Nose]
  Shape               CONICAL
  Length              15
  PowerExponent       0.5
  TipRadius           0.1
  FlatTip             false

[Body]
  Diameter            3
  Length              51.3
  BoattailLength      0
  BoattailDiameter    3
  NozzleExitDiameter  2.5

[Fins]
  Count               4
  RootChord           8
  TipChord            5.5
  Span                3.62
  Sweep               5
  AftOffset           0
  Thickness           0.157
  Section             CHAMFER
  LEChamfer           0.75
  TEChamfer           0.75
  Cant                0.2

[FinCan]
  Enabled             true
  Diameter            3.15
  Length              9.16
  AftOffset           0
  FrontBevel          1.09

[Booster]
  Enabled             true
  Diameter            3
  Length              40
  Insertion           0
  TransitionLength    1
  NoseCone            false
  NoseShape           OGIVE
  NoseLength          8
  BoattailLength      0
  BoattailDiameter    0.01
  NozzleExitDiameter  2.75

[BoosterFins]
  Count               4
  RootChord           8
  TipChord            4
  Span                4.5
  Sweep               4
  AftOffset           0
  Thickness           0.118
  Section             CHAMFER
  LEChamfer           1
  TEChamfer           0.75
  Cant                0

[BoosterFinCan]
  Enabled             true
  Diameter            3.1
  Length              8.5
  AftOffset           0
  FrontBevel          0.5

[Protuberance1]
  Type                FAIRING
  Stage               SUSTAINER
  X                   17
  Count               2
  Length              0.75
  Width               0.4
  Height              0.4
  Standoff            0

[Flight]
  Altitude            4000
  TurbulentBoundaryLayerfalse
  Roughness           0.000984252

[Motors]
  Sustainer           EAYF Sustainer - N3051
  Booster             EAYF Booster - M3140.e

[Site]
  Weather             OPENMETEO
  Latitude            35.3471
  Longitude           -117.808
  Elevation           2000
  Date                2026-09-16
  Time                17:00
  WindSpeed           0
  WindDirection       0
  CylinderRadius      7

[Launch]
  RailLength          15
  Angle               90
  Heading             0
  Coast               10
  MaxTilt             15
  MinIgnitionAltitude 0
  MaxTime             1000

[Recovery]
  ApogeeChuteDiameter 8
  ApogeeChuteCd       0.8
  MainChuteDiameter   28
  MainChuteCd         1.8
  MainChuteAltitude   2000

[Thermal]
  IncludeDescent      false

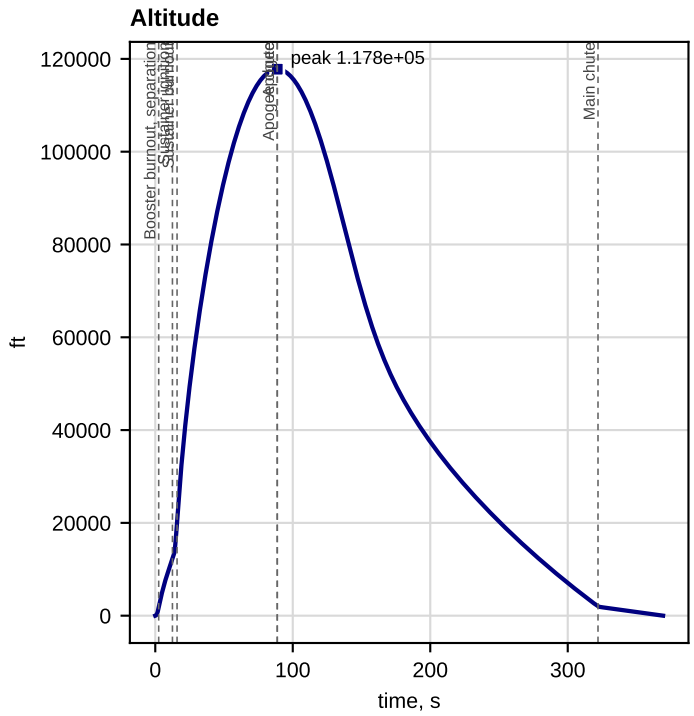
[Vibration]
  Spec                NSROC Level 2
  ThrustOscillation   15

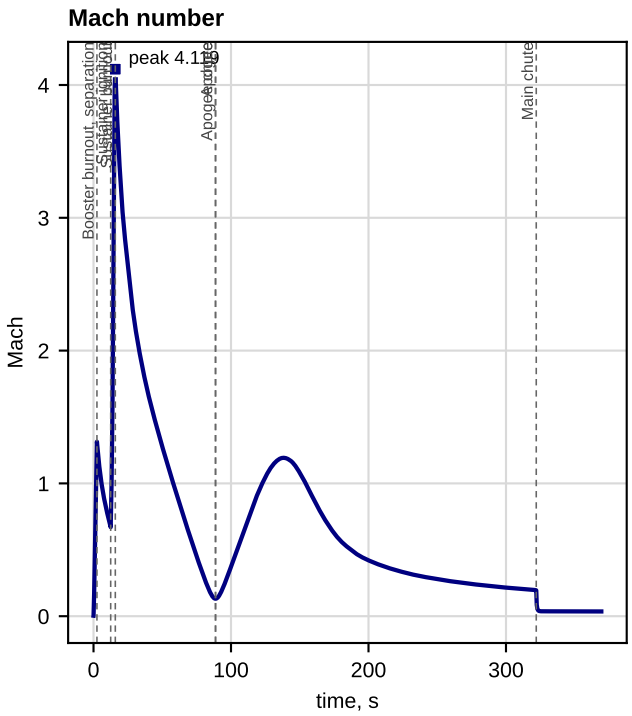
[DrySustainer]
  Mass                13.5
  CG                  42
  Inertia             4985.17
  RollInertia         37.744
  CgOffset            0.5
  ProductFin1         0.601
  Product90           0.943588
  MotorOffset         2.6

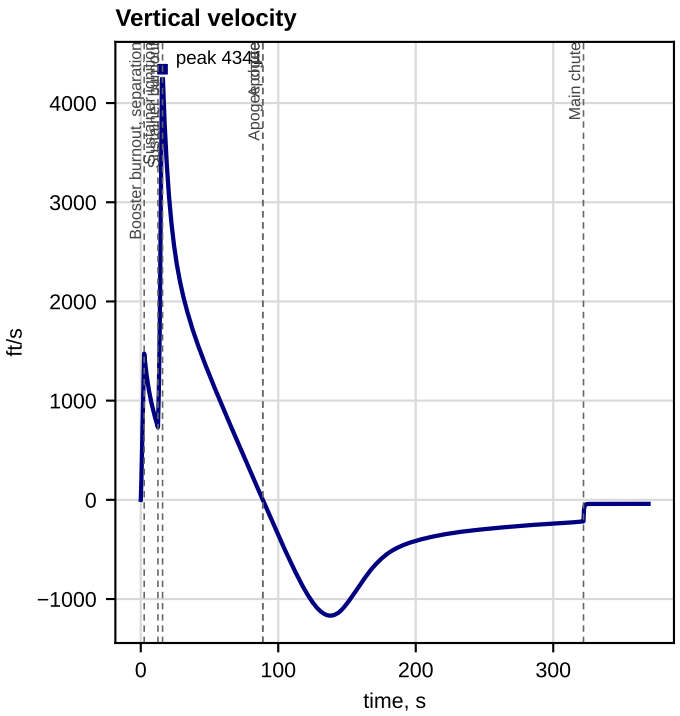
[DryStack]
  Mass                21.62
  CG                  67
  Inertia             19078
  RollInertia         63.65
  CgOffset            0.0148
  ProductFin1         1.105
  Product90           21.917
  MotorOffset         3.44

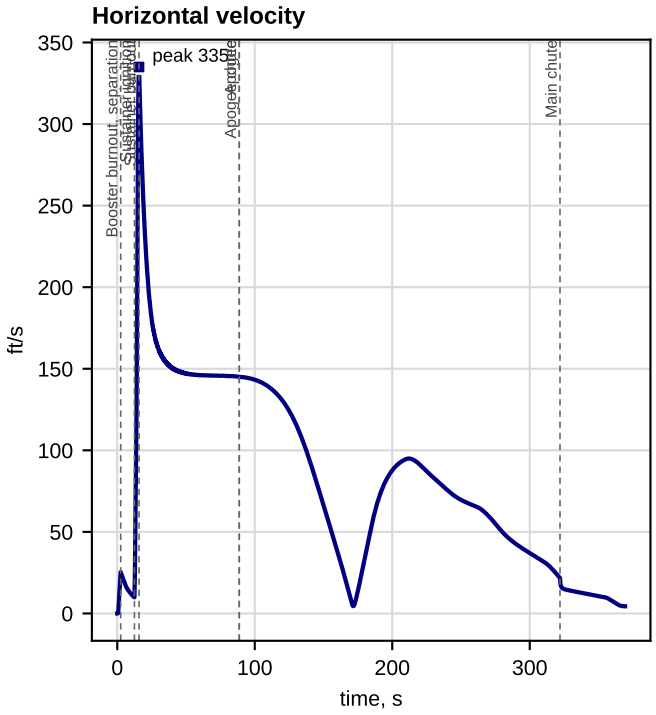
[MonteCarlo]
  Runs                0
  FailWhen            lock-in = true; resona
  ToleranceSweep      false
  AngleStd            0.5
  HeadingStd          180
  CoastStd            0.5
  BoosterImpulseStd   2
  SustainerImpulseStd 2
  BoosterMassStd      2
  SustainerMassStd    2
  CantStd             0.2
  SustainerFinAlignStd0.1
  BoosterFinAlignStd  0.1
  SustainerThrustOffsetStd0.1
  BoosterThrustOffsetStd0.2
  SustainerCgOffsetStd0.01
  BoosterCgOffsetStd  0.02
  SustainerProductStd 0.05
  BoosterProductStd   0.05
  WindSpeedStd        0
  WindDirectionStd    0
  WindDays            7
```

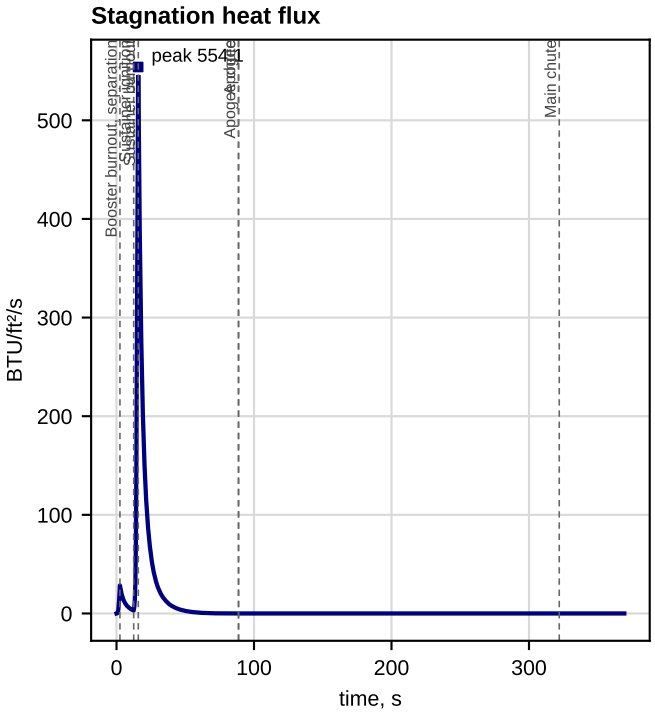
EAYF_CONFIG_1 - Flight

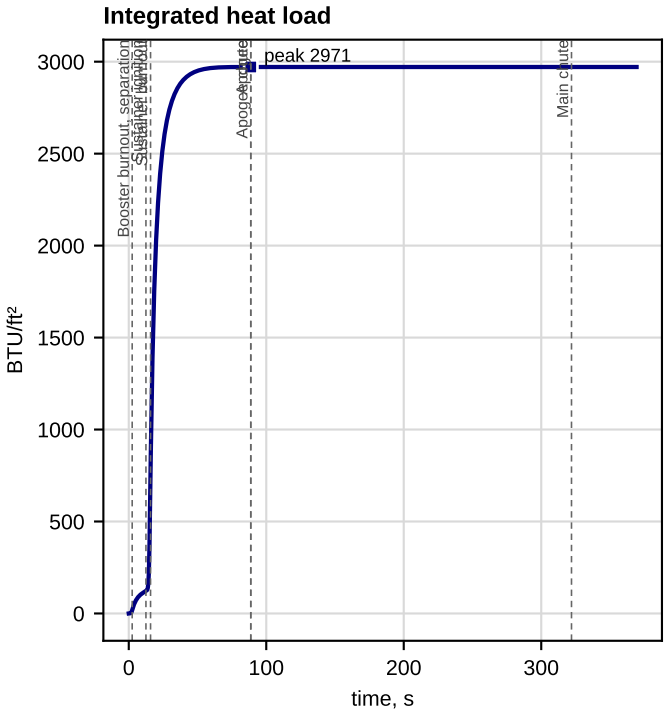






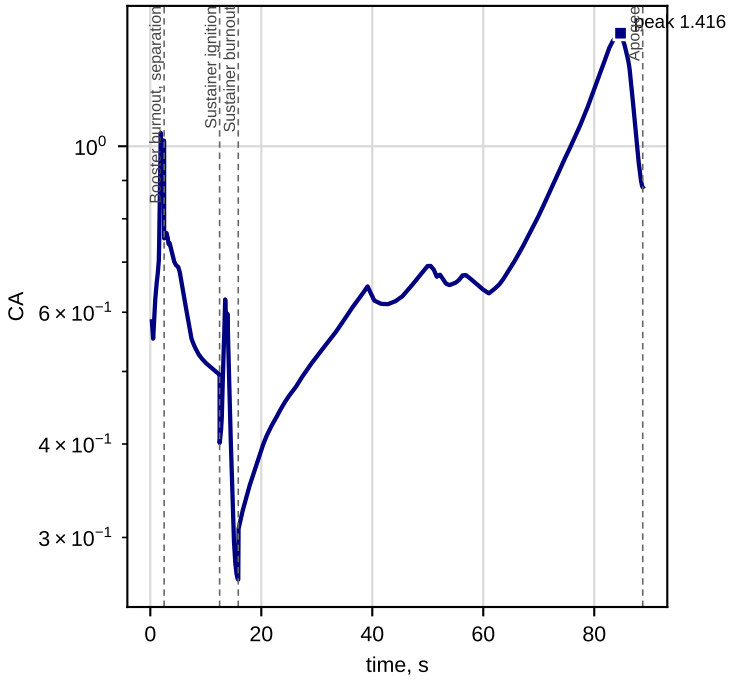




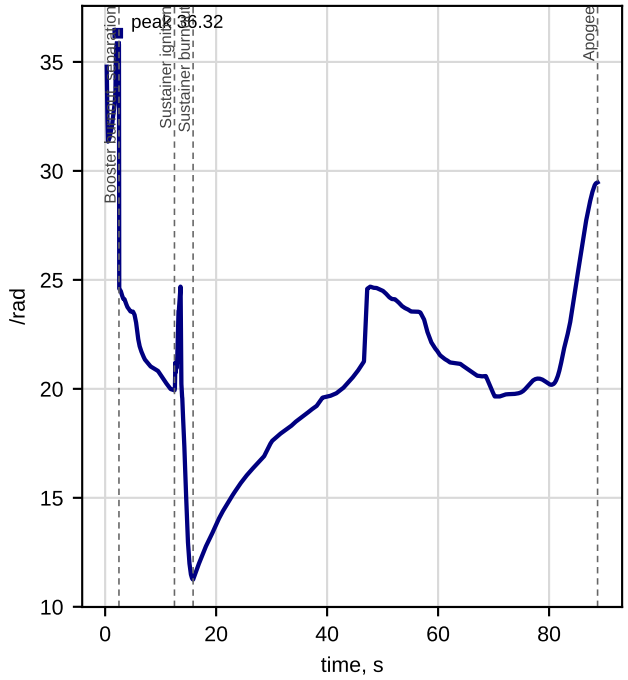


EAYF_CONFIG_1 - DATCOM aerodynamics along the flight (launch to apogee, Mach 0.1 and up)

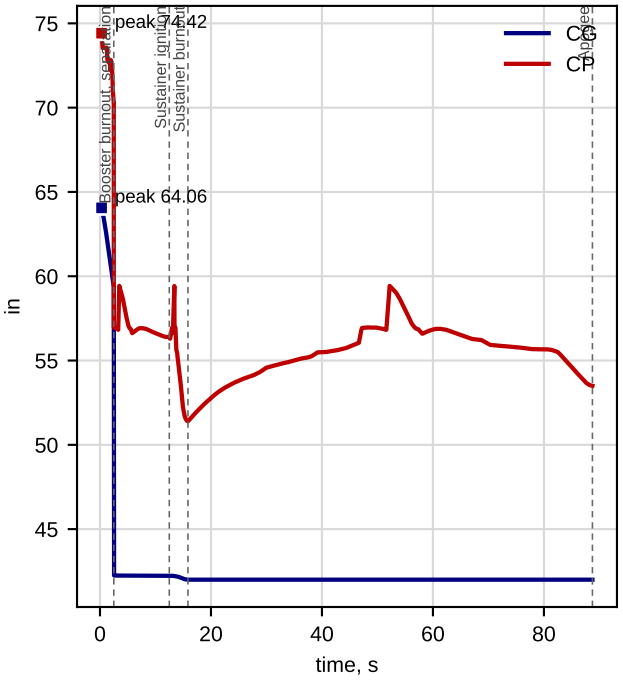
Axial force coefficient CA (log scale)



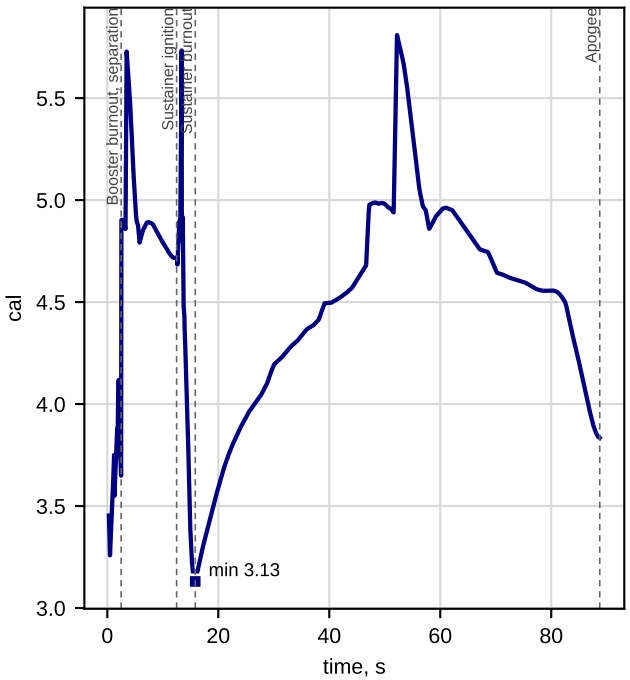
Normal force slope CNα



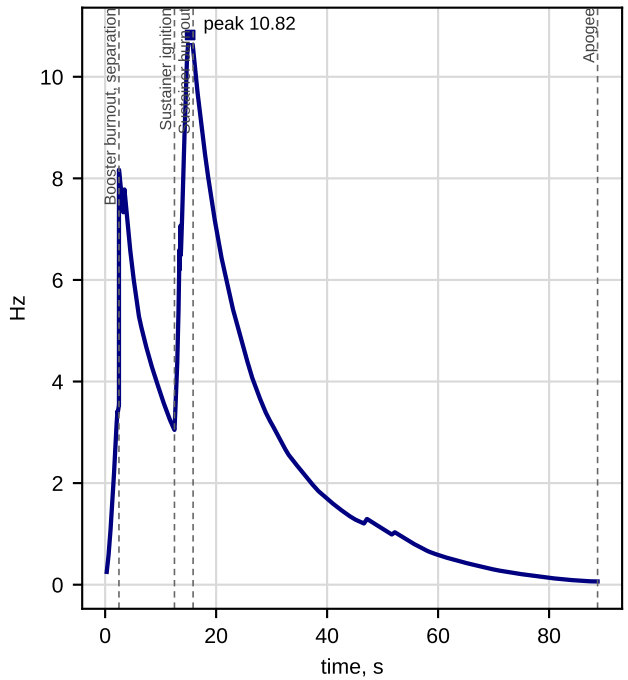
CG and centre of pressure



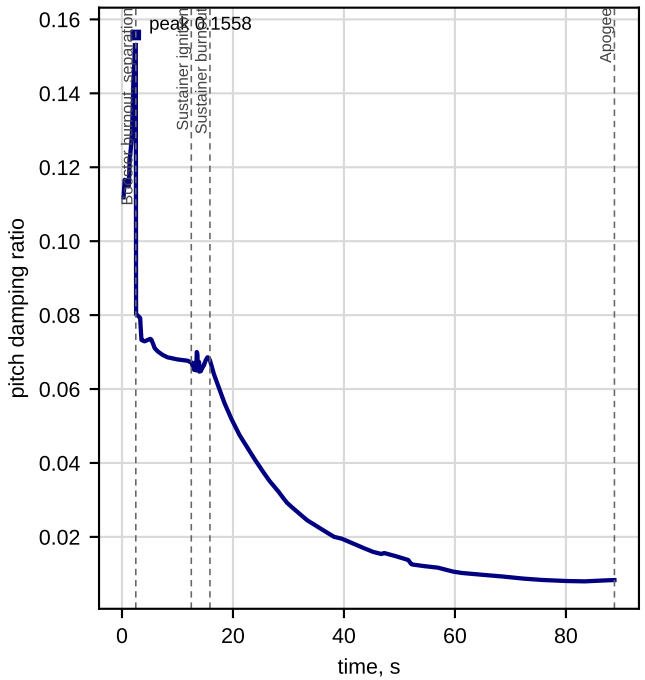
Static margin



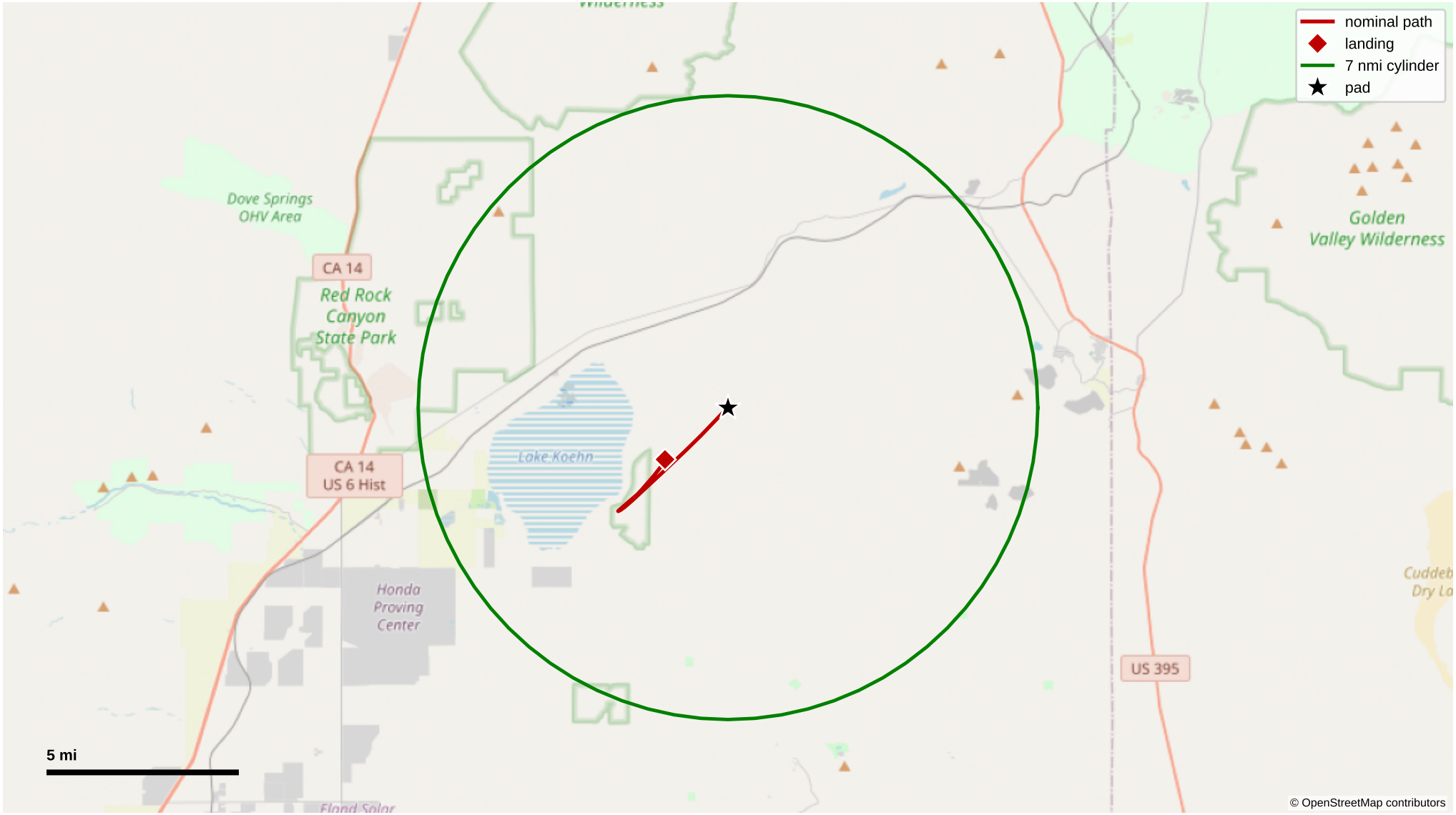
Pitch natural frequency



Pitch damping ratio



EAYF_CONFIG_1 - Ground track



EAYF_CONFIG_1 - Vibration at the avionics (model good to about ±6-10 dB; the timing and frequencies are its strength)

